Fiscal Year:	FY 2019	**Task Last Updated:**	FY 04/03/2020
PI Name:	Papier, Arthur M.D.		
Project Title:	VisualDx Augmented Intelligence Project		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
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Contact Monitor:		**Contact Phone:**	
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Task Description:	Industry Project VisualDX is the most widely used diagnostic support system in the world. VisualDX combines machine learning and vision science with structured clinical data and clinical question sets to let non-specialist healthcare providers (1) enter patient specific findings, (2) to build custom differentials, and then (3) view images, treatment recommendations and management information. The current project would create a stand-alone Android version of VisualDX that will allow for use of the platform with no required internet connection and expand its content set. The goal is to empower non-specialist providers to quickly and accurately answer clinical questions to deliver the best care.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	For NASA, this project brings a medical decision support tool that can function on deep-spaceflight missions or any other environment without access to the Internet. The VisualDx module developed for this project includes over 100 conditions on the Exploration Medical Condition List (MCL), and the infrastructure built can support the addition of more conditions and more modules in the future, including an ultrasound module currently being developed as part of another Translational Research Institute for Space Health (TRISH)-funded project (IND0009--"VisualDx Clinical Decision Support for Ultrasound"). The machine learning model and AI algorithm have been trained on spaceflight-specific dermatologic conditions that are the most common in spaceflight. As part of this project, VisualDx also improved upon NASA's MCL by expanding upon the listed diagnosis concepts to include more specific conditions. For example, while the MCL included a singular concept for "rash," VisualDx contributors were able to split that concept into over 40 specific dermatologic conditions relevant to spaceflight. For example, on the original MCL the heat rash diagnosis was not differentiated from eczema, yeast infection, or abscess. VisualDx recognized the opportunity to make this improvement early on in the grant period and was able to include much more granular concepts in the end product as a result. Further, the development of the stand-alone, space-specific version of VisualDx brings value to NASA by: - providing medical decision support in the absence of timely communication between Earth and crew members. - providing such medical decision support in a software package that is likely to run on existing computing equipment, thus requiring no additional physical storage space. - increased support and confidence in medical decision making by non-medical professionals staffing deep spaceflight missions, augmenting existing medical training protocols. The completion of this project has significant applications on Earth as well. The development of the stand-alone version enables deployment of the application in remote locations or settings with poor Internet infrastructure, making clinical decision support available to large populations previously unable to access such support. Additionally, the development to support non-specialist terminology and synonyms for medical terms makes the product more accessible to new user populations, instead of supporting only skilled clinical providers.
Task Progress:	Definitive medical care is not feasible on spaceflight missions because of a lack of in-flight medical expertise, telemedicine support, supplies, and large equipment. Therefore, definitive care may be delayed for crew members on such missions. In order to overcome these hurdles, on-board medical diagnostic systems need to either provide user-guidance capabilities or minimize training time by being exceptionally intuitive and easy to use. VisualDx is the most widely used diagnostic support system in the world. VisualDx combines machine learning and vision science with structured clinical data and clinical question sets to let non-specialist healthcare providers (1) enter patient specific findings, (2) to build custom differentials, and then (3) view images, treatment recommendations, and management information. The purpose of this project was to create a stand-alone, platform-agnostic version of VisualDx's existing application requiring no internet connection, as well as to expand the product's medical content with a focus on high-priority conditions likely to be encountered in space. The goal is to empower non-specialist providers to quickly and accurately answer clinical questions to deliver the best care. Specific project aims and objectives achieved through this project are as follows: (1) modifying the system to operate in a stand-alone environment without internet access, (2) training VisualDx image and symptom-based pattern matching technologies on diagnoses likely to occur in space, and (3) developing a space-specific content library to cover all required and optional conditions most likely to be suffered by space explorers. VisualDX has been awarded additional funding by TRISH to continue this initiative by expanding the new stand-alone product to include diagnostic support for ultrasound findings (see https://). Ultrasound is a well-accepted and well-developed modality of diagnostic medical imaging that requires minimal training for image acquisition. This project will be focused on enhancing the existing VisualDx tool's diagnostic capabilities by creating an ultrasound-specific case workup module, as well as expanding representation of ultrasound imaging to include multiple annotated images for a targeted list of conditions likely to occur in space.
Bibliography Type:	Description: (Last Updated:)